Density

We start off by considering some topological space X, endowed with a topology we will call τ_X. Then for some $A \subset X$, we have the following notion of density:

Definition 0.1. *A subset of a topological space $A \subseteq X$ is dense in X if (we propose two equivalent conditions):*

1. $\forall x \in X, \forall U_x \in \tau_X : U \cap A \neq \varnothing$
2. $\bar{A} = X$

Intuitively, no matter how close we look to a point in X, we can find a point in A, so A "covers" (perhaps an interesting adjective and intuively useful is "spans") all of X. We call this variety of dense as **everywhere dense**.

It however is possible that the closure of A is not ALL of X, but rather a subset of X. We will call this dense in a subset:

Definition 0.2. *Consider $A \subseteq Y \subseteq X$, then A is dense in Y if $Y \subseteq \bar{A}$.*

Note that we did not give the open set definition; for the purposes of further content dealing with closures is easier. Examining the condition $Y \subseteq \bar{A}$, we see that implies by definition of closure that $\bar{Y} \subseteq \bar{A}$. Furthermore, since $A \subseteq Y$, we have that $\bar{A} = \bar{Y}$. This is an equivalent condition, which will be indispensable moving forward, and we will default to this. Additionally, using the open set condition for dense in a subset requires an additional notion:

Definition 0.3. *The subspace topology is an induced topology on a subset $Y \subseteq X$, where:*

$$\tau_Y \coloneqq \{O \cap Y : O \in \tau_X\}$$

An open set relative to Y (we call this relatively open as well), is open if and only if it can be represented as the intersection of an open set $O \subseteq X$ and Y. We talked about A being dense in a subset meaning the closure of A is the closure of the subset. The closure we referred to was the closure of the sets with respect to the entire space X.

Consider the set $\mathbb{Q}$ in $\mathbb{Q}$ versus $\mathbb{Q}$ in $\mathbb{R}$. Then the closure of $\mathbb{Q}$ in the first case is $\mathbb{Q}$ itself; whereas, the closure in $\mathbb{R}$, is $\mathbb{R}$. Then:

Lemma 0.1. *The closure of A with respect to Y, which we will denote as $\overline{A}^Y$ is the same as $\bar{A} \cap Y$.*

Proof. We know that the definition of closure with respect to X is:

$$\bar{A} \coloneqq \bigcap \{C : C \text{ closed} \land A \subseteq C\}$$

Then for all such C which are closed subsets of Y, there exists some $B \subseteq X$ such that B is closed and $C = B \cap Y$. Then:

$$\bar{A}^Y = \bigcap C = \bigcap (B \cap Y) = \left(\bigcap B\right) \cap Y = \bar{A} \cap Y$$

□

Lemma 0.2. *For A dense in a subset Y, another equivalent form of the statement is $\bar{A}^Y = Y$.*

Proof. We first prove that $\bar{A} = \bar{Y}$ yields the equivalent statement. Simply take the intersection with Y for both sets such that:

$$\bar{A} \cap Y = \bar{A}^Y = \bar{Y} \cap Y = Y$$

Now if we have that $\bar{A}^Y = Y$, we can write it as $\bar{A} \cap Y = Y$. Thus:

$$Y = \bar{A} \cap Y \subseteq \bar{A} \implies \bar{Y} \subseteq \bar{A}$$

By subset properties once more, we get $\bar{A} = \bar{Y}$. $\square$

Notice how in Definition 0.2, we required $A \subseteq Y$. We can actually weaken this condition to simply a nonempty intersection:

Definition 0.4. *A is* ***somewhere dense*** *if there exists some open $Y \subseteq X$ such that $\overline{A \cap Y} = \bar{Y}$*

Consider some open set Y such that $O \coloneqq A \cap Y$ and $A \cap Y \neq \varnothing$. Then $O \subseteq Y$, so if O is dense in Y:

$$\bar{O}^Y = Y \iff \bar{O} = \bar{Y} \iff \overline{A \cap Y} = \bar{Y}$$

This property of somewhere dense means part of A can span some subset, not necessarily the whole set. We give a few facts without apology:

1. A subset A is everywhere dense (we will refer to this as simply dense) if and only if somewhere dense for all open subsets Y of X
2. A subset A is somwhere dense if and only if $\text{int}(\bar{A}) \neq \varnothing$

We introduce the notion, jumping off of somewhere dense, of nowhere dense.

Definition 0.5. *A subset A is nowhere dense if and only if it is NOT somewhere dense.*

Notice by the (1) in the previous list, not dense **DOES NOT** imply nowhere dense. If a set is not dense, then there exists some Y in which it is not somewhere dense (but the lack of somewhere density is not necessarily true for all such open subsets).

However, a nowhere dense subset is not dense. By (2) we see that a subset A is nowhere dense if and only if $\text{int}(\bar{A}) = \varnothing$. This will be the main criterion for proving nowhere density as we will see with examples such as the Cantor set. We state an additional corollary:

Corollary 0.1. *If A is somewhere dense for some open $Y \subseteq X$, then $\overline{A \cap Y} = \bar{A} \cap \bar{Y}$*

We already have the inclusion that $\overline{A \cap Y} \subseteq \bar{A} \cap \bar{Y}$ (which is true for all sets). We prove the reverse inclusion. We see that:

$$\bar{Y} = \overline{A \cap Y} \wedge A \cap Y \subseteq A \implies \bar{Y} \subseteq \bar{A}$$

Thus $\bar{A} \cap \bar{Y} = \bar{Y} = \overline{A \cap Y} \supseteq \bar{A} \cap \bar{Y}$.

The Cantor Set

Properties

The Cantor Set is a nowhere dense, perfect set constructed by repeatedly removed the middle 1/3 interval of each interval. We denote each step of this iterative process as C_n where for all $I \subset C_{n-1}$, we remove their middle 1/3 portion, creating C_n.

We first note that the Cantor set is nonempty. Since for all n, we have that $C_{n+1} \subseteq C_n$ and because $[0,1]$ is closed and bounded (compact), by Cantor intersection property we have that their intersection, which is C, is nonempty.

An interesting property of the Cantor Set is that it is uncountable. For this, we examine a representation of the elements of the Cantor Set, called the ternary expansion:

$$\forall x \in C : x \coloneqq \sum_{i \in \mathbb{N}} \frac{2\sigma_i}{3^{i+1}}$$

where σ_x is a sequence in $\{0,1\}^{\mathbb{N}}$. This is called the ternary representation because it is in base 3. If a representation of a number in $[0,1]$ MUST contain a 1 in it, then that tells us it is in a middle 1/3 interval.

Based on its place in the decimal, such a ternary expansion tells us that in order to reach the number, we must pass through the interval $[1/3^n, (2/3)^n)$ (and endpoints remain in the Cantor Set), so it belongs in the middle interval.

This ternary expansion gives us an injection $f : \{0,1\}^{\mathbb{N}} \to C$. If $f(\sigma_1) = f(\sigma_2)$, then:

$$f(\sigma_1) = f(\sigma_2) \implies \sum_{i \in \mathbb{N}} \frac{2\sigma_{1i}}{3^{i+1}} = \sum_{i \in \mathbb{N}} \frac{2\sigma_{2i}}{3^{i+1}}$$

$$\implies \sum_{i \in \mathbb{N}} \frac{2(\sigma_{1i} - \sigma_{2i})}{3^{i+1}} = 0$$

Since 0 has a unique ternary expansion which is $0.00000..._3$ itself, then this means $\forall i \in \mathbb{N} : \sigma_{1i} = \sigma_{2i}$, proving equality of the sequences and injectivity of f.

Now because $\{0,1\}^{\mathbb{N}} \simeq \mathbb{R}$, then $\mathbb{R} \lesssim C$, so the Cantor set is uncountable (which will be an interesting result later on).

We note that each time the iterative process doubles the amount of intervals, so each C_n can be covered by 2^n closed balls of radius $3^{-n}/2$. This fact will be quite important when we examine integration over the indicator function on C. We first prove that the Cantor set is nowhere dense.

Lemma 0.3. *The Cantor Set is nowhere dense.*

First since each C_n is a closed set, then the construction of C_{n+1} involves removing only OPEN middle 1/3 sets, so C_{n+1} is also a finite union of 2^n closed sets, thus closed. C is also closed because it is the arbitrary union of closed sets C_n, so $\bar{C} = C$.

A point is in the interior if of a set A:

$$\exists \epsilon > 0 : B(x, \epsilon) \subseteq A$$

So we aim to prove the negation of this for all such points in the Cantor set. For all x in the Cantor set, since $x \in \cap_{n \in \mathbb{N}} C_n$, then $\forall n \in \mathbb{N} : x \in C_n$. We can find the following:

$$\forall \epsilon > 0, \exists n \in \mathbb{N} : \frac{1}{3^n} < 2\epsilon$$

Then we have that x belongs to some interval $I_{n'}$ in C_n. Because each interval is covered by a closed ball of radius $1/(2 \cdot 3^n)$ centered at the midpoint (which may be x or not), then the ball $B(x, \epsilon)$ certainly exceeds the interval.

(If x is the midpoint, this follows from the covering. If x is not the midpoint, then the ball $B(x, 1/(2 \cdot 3^n))$ already falls outside the interval, so we can extend it to the ϵ-ball by subset properties). Thus:

$$\forall \epsilon > 0, \exists y \notin C : y \in B(x, \epsilon)$$

Thus the Cantor set has no interior points and is nowhere dense.

Lemma 0.4. *The Cantor Set is perfect, which means it is closed and has no isolated points.*

Given some $x \in C$ and some $\epsilon > 0$, the previous lemma gave us the existence of $n \in \mathbb{N}$, for which x belongs to an interval in C_n. Once again by case work, we can show that at least one endpoint of the interval different from x should fall in the ball $B(x, \epsilon)$ and since endpoints remain in C, we have shown that no point is isolated:

$$\forall x \in C, \forall \epsilon > 0 : (B(x, \epsilon) \setminus \{x\}) \cap C \neq \varnothing$$

Integrability

We now examine the following function:

$$1_C \coloneqq \begin{cases} 1 \ : \ x \in C \\ 0 \ : \ x \notin C \end{cases}$$

We aim to show this function is Riemann integrable. We first show that it is discontinuous at all points within the Cantor set (via limsup liminf criterion):

$$\forall x \in C, \forall \delta > 0 : \sup_{z:|z-x|<\delta} f(x) = 1 \wedge \inf_{z:|z-x|<\delta} f(x) = 0$$

The infimum part is true because of Lemma 0.3 (nowhere dense), which tells us that for any δ-ball around x, we can always find some point outside the Cantor set within the ball. The supremum can also be supported by the perfect + nowhere dense properties of C (if we consider the punctured open ball, perhaps to disprove existence of a limit). In that case, we have a discontinuity of second kind at each $x \in C$ as we can show both left and right limits do not exist.

Then since the above condition is true for all δ:

$$\lim_{\delta \to 0+} \sup_{z:z \in B(x,\delta)} f(x) = 1 > \lim_{\delta \to 0+} \inf_{z:|z-x|<\delta} f(x) = 0$$

which tells us that f is not continuous at all $x \in C$. So now we have a function that is nowhere dense, with an uncountable number of discontinuities, that too of the second kind YET is Riemann integrable. Then:

1. Cardinality of the set of discontinuities while sufficient is too strong of a condition for Riemann integrability
   (a) Thomae's function in $[0, 1]$ showed that we can have countably many discontinuities (but of the first kind as limit exists at all rationals)
2. Type of discontinuities faces the same issue as well (1_C has discontinuities of the second kind)

Lemma 0.5. *The function 1_C is bounded and Riemann integrable on $[0, 1]$ and $\int_0^1 1_C dx = 0$.*

We note that by the Darboux definition, the lower integral over any partition is always 0. So we aim to show that the upper integral also converges to 0. We define the following:

$$\Pi_n \coloneqq \{I_n \cup M_n : I_n \subseteq C_n\}$$

where I_n are the existing intervals after removing the middle 1/3 intervals denoted as M_n. Then:

$$\forall \Pi_n : U(f, \Pi_n) = \sum_{I_{n_i} \in I_n} 1 \cdot \Delta I_{n_i} + \sum_{M_{n_j} \in M_n} 0 \cdot \Delta M_{n_j}$$

This is because the supremum on I_n is always 1 (the endpoints are in C) and the supremum on M_n is 0 (it contains no points in the Cantor set). Then:

$$U(f, \Pi_n) = 1 \cdot \left(\frac{2}{3}\right)^n \implies \inf U(f, \Pi_n) = \lim_{n \to 0} U(f, \Pi_n) = 0$$

Thus the upper integral and lower integral agree at the value 0. The $(2/3)^n$ come from the fact that there are 2^n closed balls of radius $1/(2 \cdot 3^n)$ that cover all of C_n. Notice that the "length" of the Cantor set (the set of discontinuities in this case) goes to 0.

Measure

Definitions

We never officially introduced the notion of length. We first say that the length of any OPEN interval (a, b), (denoted by l) in $\mathbb{R}$ is $b - a$. Length we will consider to be a map $l : \mathcal{P}(\mathbb{R}) \to [0, \infty]$ (range in extended reals). We now introduce length of a set as the following:

Definition 0.6. *The outer measure of a set A is via covering sequences of intervals $\{I_n\}_{n \in \mathbb{N}}$:*

$$|A| \coloneqq \inf \left\{ \sum_{n \in \mathbb{N}} l(I_n) : A \subseteq \bigcup_{n \in \mathbb{N}} I_n \right\}$$

We note that the measure of the empty set is 0. The intuition behind this is consider all possible covers for A, then the smallest possible cover (the one that "hugs" or matches A's shape the "tightest"), can best approximate its length. The idea of measure will give us the equivalent criterion for integrability, which we will cover later on.

Another way we can restate this property is as the following:

$$\forall \epsilon > 0, \exists \{I_n\}_{n \in \mathbb{N}} : \sum_{n \in \mathbb{N}} I_n - |A| < \epsilon$$

First we examine some properties of measure:

Lemma 0.6. *The measure of a finite set of points is* 0.

Proof. Let our finite set of points be $A \coloneqq \{x_0, ..., x_m\}$. Then we can cover A by the following sequence:

$$I_n \coloneqq \begin{cases} (x_n - \epsilon, x_n + \epsilon) \ : \ n \leqslant m \\ \varnothing \ : \ n > m \end{cases}$$

$$|A| \leqslant \sum_{n \in \mathbb{N}} l(I_n) = 2m\epsilon$$

Because the length of an interval is always non-negative, we note that the measure also must be non-negative. If we assume some set were to have measure less than 0, then by properties of infimum being adherent, there would exist a sequence of covering sets such that their length is negative (not possible). Since ϵ is an arbitrary number greater than 0, we can find any value of ϵ less than a possible value of $|A|$, thus squeezing the measure so that $|A| = 0$. $\square$

This proof and non-negativity of the metric tells us that it too is a map from $\mathcal{P}(\mathbb{R}) \to [0, \infty]$. We present a stronger version of Lemma 0.6:

Lemma 0.7. *The measure of a countable set of points is 0.*

Proof. Consider a similar setup as Lemma 0.6, with $A \coloneqq \{x_0, x_1, ...\}$ and a specific $\{I_n\}_{n\in\mathbb{N}}$ defined as:

$$I_n \coloneqq (x_n - 2^{-n-1}, x_n + 2^{-n-1})$$

Then:

$$|A| \leqslant \sum_{n\in\mathbb{N}} I_n = \sum_{n\in\mathbb{N}} \frac{\epsilon}{2^n} = 2\epsilon$$

The result comes from the sum of the geometric series of $(1/2)^n$. Once again we have a similar squeeze argument, showing $|A| = 0$. This is of interest to us as it shows that Thomae's function in $[0, 1]$ has a set of discontinuities of zero measure. □

So far we have only talked about measure in terms of open intervals I_n, however we now show that it is equivalent for closed intervals. We prove the following two Lemmas:

Lemma 0.8. *If $A \subseteq B$, then the measure of A is less than or equal to the measure of B.*

Proof. Consider some sequence $\{I_n\}_{n\in\mathbb{N}}$ such that the union of the intervals cover B. Then:

$$A \subseteq \bigcup_{n\in\mathbb{N}} I_n \implies |A| \leqslant \sum_{n\in\mathbb{N}} l(I_n)$$

By Definition 0.6 (infimum properties), we get that $|A|$ is a lower bound for all such sums, thus $|A| \leqslant |B|$. □

Lemma 0.9. *The measure of an open interval (a, b) is equal to the measure of the closed interval $[a, b]$.*

Proof. We defined the measure of the open interval (a, b) as its length $b - a$. Then consider:

$$\forall \epsilon > 0 : [a, b] \subseteq (a - \epsilon, b + \epsilon)$$

By Lemma 0.8, we see that:

$$|[a, b]| \leqslant |(a - \epsilon, b + \epsilon)| = (b - a) + 2\epsilon$$

Since ϵ is arbitrary, this tells us that $|[a, b]| \leqslant b - a = |(a, b)|$. Lemma 0.8 also yields the converse:

$$(a, b) \subseteq [a, b] \implies |(a, b)| = b - a \leqslant |[a, b]|$$

Thus $|[a, b]| = |(a, b)| = b - a$. This allows us to define measure in terms of closed intervals $\bar{I_n}$ as well as they are of equivalent length. □

Lemma 0.8 allows us to further argue that the measure of the set of discontinuities (or the measure of the Cantor set) for the function in Lemma 0.5 is 0:

$$\forall n \in \mathbb{N} : C \subseteq C_n \implies |C| \leqslant |C_n|$$

We know that 2^n closed balls of radius $1/(2 \cdot 3^n)$ cover the set so that $C_n \ \cup I_n$. The measure of each C_n can be calculated by the measure of each interval I_n, for which we require the lemma:

Lemma 0.10. *Outer measure is subadditive so that:*

$$\forall n \in \mathbb{N} : \left| \bigcup_{n\in\mathbb{N}} A_n \right| \leqslant \sum_{n\in\mathbb{N}} |A_n|$$

This is a triangle inequality-"ish" property, while not exactly as at one side we have set unions and at the other side we have addition. Let I_n be a sequence of intervals covering its respective A_n, and I_{ni} denoting an element in the sequence covering A_n. Then:

$$\bigcup_{n\in\mathbb{N}} A_n \subseteq \bigcup_{n\in\mathbb{N}} I_n = \bigcup_{n\in\mathbb{N}} \bigcup_{m\in\mathbb{N}} I_{nm}$$

Since each I_n is a cover for A_n, we have that:

$$\forall n \in \mathbb{N} : |A_n| \leqslant \sum_{m\in\mathbb{N}} l(I_{nm}) \implies \sum_{n\in\mathbb{N}} |A_n| \leqslant \sum_{n\in\mathbb{N}} \sum_{m\in\mathbb{N}} l(I_{nm})$$

Now for all $\epsilon > 0$, by the equivalent definition of measure (derived from supremum), we can find sequences I_n, such that:

$$\sum_{m\in\mathbb{N}} l(I_{nm}) \leqslant \sum_{n\in\mathbb{N}} |A_n| + \frac{\epsilon}{2^n}$$
$$\implies \sum_{n\in\mathbb{N}} \sum_{m\in\mathbb{N}} l(I_{nm}) \leqslant \sum_{n\in\mathbb{N}} |A_n| + 2\epsilon$$

We now need to show the existence of a covering sequence for $\cup A_n$. Consider the sequence where the indices nm add up to a specific number (starting from 0), so that we get:

$$\{I_{00}\}, \{I_{01}, I_{10}\}, \{I_{11}, I_{20}, I_{02}\}...$$

Then $\forall \epsilon > 0$, there exists a sequence as defined above such that:

$$\left| \bigcup_{n\in\mathbb{N}} A_n \right| \leqslant \sum_{n\in\mathbb{N}} \sum_{m\in\mathbb{N}} l(I_{nm}) \leqslant \sum_{n\in\mathbb{N}} |A_n| + 2\epsilon$$

Since it is true for all epsilon, then we arrive at out result. This however only proves subadditivity (which in our case to prove the zero measure of the Cantor set would be enough, but we can prove something stronger).

Lemma 0.11. *For a sequence of disjoint open intervals I_n, we have that:*

$$\left| \bigcup_{n\in\mathbb{N}} I_n \right| = \sum_{n\in\mathbb{N}} |I_n|$$

I_n is covering sequence for itself (and also we can infer by Lemma 0.10) the inequality:

$$\left| \bigcup_{n\in\mathbb{N}} I_n \right| \leqslant \sum_{n\in\mathbb{N}} |I_n|$$

By the definition of infimum and measure, we also have that:

$$\forall \epsilon > 0, \exists \{I'_n\}_{n\in\mathbb{N}} : \sum_{n\in\mathbb{N}} |I'_n| \leqslant \left| \bigcup_{n\in\mathbb{N}} I_n \right| + \epsilon$$

In order to prove this, we need to prove disjoint additivity in the finite sense, such that:

$$\left|\bigcup_{n=0}^{N} I_n\right| = \sum_{n=0}^{N} l(I_n)$$

Once again by definition of measure, the forward inequality holds. We prove the reverse inequality by induction. Our base case is when $N = 0$, in which it follows from the definition of length and measure. Now for $N + 1$, we once again have by definition:

$$\left|\bigcup_{n=0}^{N+1} I_n\right| \leqslant \sum_{n=0}^{N+1} l(I_n)$$

On the other hand:

$$\sum_{n=0}^{N+1} l(I_n) = \sum_{n=0}^{N} l(I_n) + l(I_{n+1}) = \left|\bigcup_{n=0}^{N} l(I_n)\right| + l(I_{n+1})$$

Now let I'_n be an arbitrary covering sequence for the union of I_n up to $N + 1$, the define:

$$\begin{aligned} J_n &\coloneqq I'_n \cap I_{N+1} \\ J'_n &\coloneqq I'_n \cap (\mathbb{R} \setminus I_{N+1}) \end{aligned}$$

Because we have the condition of disjointness, we are able to separate the cover I'_n into these two sequences such that $I'_n = J_n \cup J'_n$. We note that J_n exclusively covers I_{N+1} and does not intersect with any of the other sets, while J'_n exclusively covers the rest of the union. Therefore:

$$\sum_{n\in\mathbb{N}} |I'_n| = \sum_{n\in\mathbb{N}} (|J_n| + |J'_n|) \geqslant \left|\bigcup_{n=0}^{N} I_n\right| + |I_{N+1}|$$

Once again since it is a covering sequence for the union up to $N + 1$, for any such sequence, the right hand side is a lower bound, so by infimum properties, we have that:

$$\left|\bigcup_{n=0}^{N+1} I_n\right| \geqslant \left|\bigcup_{n=0}^{N} I_n\right| + |I_{N+1}|$$

thereby proving equality.

Now returning to the Cantor set, by Lemma 0.11, we see that since each C_n is a closed union of disjoint intervals of length 3^{-n}:

$$\forall n \in \mathbb{N} : |C_n| = \left(\frac{2}{3}\right)^n \implies \forall n \in \mathbb{N} : |C| \leqslant \left(\frac{2}{3}\right)^n$$

By a similar argument to Lemma 0.7, we see that $n \to 0 \implies |C| = 0$. If we say $|C| = \epsilon > 0$, then $\exists n \in \mathbb{N} : (2/3)^n < \epsilon$, which would mean $|C_n| < |C|$ (contradiction). Now the reader may think that the Cantor set is nowhere dense, so nowhere dense sets have zero measure; however, that is false. First consider the converse:

$$\text{zero measure} \not\Rightarrow \text{nowhere dense}$$

For this consider Thomae's function in the interval $[0, 1]$. The discontinuities (of the first kind) are at $\mathbb{Q}\cap[0, 1]$. Since this is a countable set, we have that its measure is 0. However:

$$\overline{\mathbb{Q} \cap [0, 1]} = \bar{\mathbb{Q}} \cap [0, 1] = [0, 1]$$

So the set of discontinuities is dense in all of $[0, 1]$ (this also shows rationals are somewhere dense within the interval by Defn 0.4). For the converse, we introduce a more intricate idea of the Fat Cantor set.

The Fat Cantor Set

The Fat Cantor Set or rather the family of such sets are examples of sets which are nowhere dense but DO NOT have measure of 0. We consider a sequence $\{a_n\}_{n\in\mathbb{N}} \in (0,1)^{\mathbb{N}}$, such that:

$$\sum_{n\in\mathbb{N}} a_n < \infty$$

We can now calculate the measure of each stage of the Fat Cantor set:

$$|F_n| := \prod_{n\in\mathbb{N}} (1-\alpha_n)$$

To prove that this product does not diverge or converge to 0, we can rewrite it in terms of sums as the following:

$$\prod_{n\in\mathbb{N}}(1-a_n) = e^{\ln\left(\prod_{n\in\mathbb{N}}(1-a_n)\right)} = e^{\sum_{n\in\mathbb{N}}\ln(1-a_n)}$$

Now the convergence of the product then depends on the convergence of the sum of ln of its terms. If the sum converges to some value L, then the product converges to sum non-zero positive value (it only converges to zero if the sum diverges to $-\infty$). Now observe the power series for $\ln(1-x)$ around $x=0$:

$$ln(1-x) < P_2(x) := 0 - x - \frac{x^2}{2}$$

The less than inequality comes from the concavity of ln (any tangent line/graph falls above it), so we see that $\ln(1-x) < -x$ for all $x \in (0,1)$. Thus:

$$sum_{n\in\mathbb{N}} \ln(1-a_n) < \sum_{n\in\mathbb{N}} -a_n$$

and because we are given that the sum of $a_n's$ converge, then the sum on the right hand side also converges. Thus as $n\to\infty, |F_n| \to L > 0$, so the Fat Cantor set's measure is NONZERO. We now show that the Fat Cantor Set is nowhere dense. We see that the length of each interval $I_m^{(n)}$ in the nth stage can be bounded above:

$$|I_m^(n)| = \frac{1}{2^n}\prod_{n\in\mathbb{N}}(1-a_n) \leqslant \frac{1}{2^n}$$

So given some point $x \in F$, then given some $\epsilon > 0$, we can always find $n \in \mathbb{N}$ such that $2^{-n} < \epsilon$, which means:

$$\exists y \in B(x,\epsilon) : y \notin F_n \implies y \notin F$$

So the Fat Cantor set is an example of a set that is nowhere dense yet not zero measure. Thus the two notions are separate (we showed the set of discontinuities in Thomae's function are dense but zero measure).